

Mahesh Chitakoti

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Summary

Full Stack Engineer with 2.5+ years of experience building responsive **React/Next.js** frontends and scalable **Node.js/FastAPI** microservices. Demonstrated track record of slashing API latency by **60%** via **Redis caching and PostgreSQL indexing**, with hands-on expertise in automated testing, real-time event streaming, and cloud deployments on **AWS**.

Technical Skills

Frontend Engineering: React.js, Next.js, TypeScript, JavaScript (ES6+), Redux Toolkit, Tailwind CSS, HTML5, CSS3, Responsive Design

Languages & Backend: Node.js, Express.js, NestJS, Python, FastAPI, SQL, AsyncIO, REST APIs, Microservices, WebSockets, MQTT

Databases & Caching: PostgreSQL, MongoDB, Redis (Queues, Caching), pgvector, B-tree/Compound Indexing, Connection Pooling

Cloud & DevOps: AWS (EC2, S3), Docker, GitHub Actions CI/CD, Nginx, Linux, Vercel

Testing & Tooling: Jest, React Testing Library, Postman (API Testing), Git, Webpack, Vite

AI & Search Integration: Vector Search (Qdrant, pgvector), Hybrid Retrieval (BM25), Semantic Caching, LLM API Integration (Gemini, OpenAI)

Experience

AI Developer – SahioNext Technology Pvt. Ltd.

Oct 2025 – Apr 2026

Remote

- Architected an end-to-end full-stack media platform using **FastAPI, React.js, and PostgreSQL**, decoupling asynchronous workloads for **10,000+ audio sessions** via **Redis queues and background workers**.
- Built responsive operational telemetry dashboards in **React.js** and **Redux Toolkit** tracking live token spend, rate limits, and pipeline stages, reducing issue triage time from hours to minutes.
- Engineered full-duplex **WebSocket and MQTT** ingestion pipelines with **exponential backoff, dead-letter queues (DLQ), and automatic session recovery**, maintaining resilient real-time streaming across high-latency clients
- Integrated high-performance semantic search microservices in **Qdrant** (<300ms p95 latency) across **10,000+ indexed chunks**, enabling rapid CEFR language proficiency grading within user review workflows.

Full Stack Developer (Contract) – PearlThoughts

Feb 2025 – Sep 2025

Remote

- Engineered multi-tenant backend automation microservices using **NestJS, TypeScript, and Google Workspace APIs**, orchestrating **1,000+ daily business workflows** and reducing manual client data-entry bottlenecks.
- Built distributed workflow pipelines with **exponential backoff, idempotent webhook handlers, and rate limiting**, ensuring resilient, idempotent job execution and preventing duplicate transaction processing across tenants.
- Developed automated verification harnesses and isolated **Docker autograder environments** for enterprise **Coursera course launches**, preventing runtime execution regressions prior to production rollout.

Full Stack Developer – The DevOps Team

Oct 2023 – Jan 2025

Bangalore, India

- Optimized high-traffic **Node.js** and **Express.js** APIs using **Redis caching, PostgreSQL B-tree indexing, and connection pooling** (reducing latency from **800ms to 320ms**), writing **Jest** and **Postman integration tests** to maintain high test coverage.
- Developed candidate-job recommendation workflows utilizing multi-attribute compound indexing in **PostgreSQL**, cutting search filter latency and improving relevant role matching accuracy by **22%**.
- Built **React.js** and **Next.js** onboarding experiences with integrated **A/B testing**, lifting user signup conversion rates by **18%**.

Projects

WorkBridge – Full-Stack Project Collaboration Platform

May 2026 – Present

React.js, Next.js, Node.js, Express.js, TypeScript, PostgreSQL, pgvector, Redis, OAuth, Tailwind CSS | Live Project

- Designed and implemented a decoupled microservices platform featuring a **Node.js/Express API gateway** for session/auth handling and a **Python/FastAPI microservice** for vector processing.

- Built core REST APIs with **PostgreSQL and Redis**, enforcing role-based access control (RBAC) and OAuth2 authentication to achieve **150ms p95 latency in k6 benchmark load tests**.
- Developed dynamic client-side interfaces in **React.js, Next.js, and Tailwind CSS** featuring responsive project boards, profile matching filters, and real-time interaction feeds.

Certifications

Oracle Agentic AI Certified Foundations Associate	– Oracle	Aug 2026
Oracle AI Vector Search Certified Professional	– Oracle	Sep 2025
Machine Learning I	– Columbia University	Aug 2025
Oracle Cloud Infrastructure 2024 Generative AI Certified Professional	– Oracle	Aug 2024

Education

Adichunchanagiri Institute of Technology (VTU)	Chikkamagaluru, KA, India
<i>Bachelor of Engineering in Mechanical Engineering</i>	Aug 2023